AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM256
Cyber Security

Final Examination Key
Fall Semester 2025/2026

Date: 13/Jan./2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed: 3 Hours**

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **A calculator is allowed.**

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PART 1: Multiple Choice, True or False questions [15 Marks]

Question 1: Choose the correct answer for the following questions [15 Marks]:

1. Authentication is the process of:
   a) Determining what actions, a user can perform
   b) Verifying a user's identity
   c) Encrypting system files
   d) Monitoring log files

2. Which term refers to the process of identifying critical organizational assets and their associated threats?
   a) Risk analysis
   b) System audit
   c) Patching
   d) Encryption

3. Which of the following best describes multi-factor authentication (MFA)?
   a) Using multiple passwords
   b) Using different devices to access systems
   c) Combining two or more independent authentication methods
   d) Reusing passwords across services

4. The most effective way to mitigate insider threats is to:
   a) Block all USB ports
   b) Use employee monitoring only
   c) Combine technical controls with employee awareness programs
   d) Reduce system privileges for all users

5. What is the biggest challenge in symmetric encryption?
   a) Speed of encryption
   b) Key distribution and management
   c) Use of mathematical complexity
   d) Licensing requirements

6. What is the function of a digital signature?
   a) To encrypt data for confidentiality
   b) To verify data authenticity and integrity
   c) To increase storage efficiency
   d) To enable faster communication

7. The principle of least privilege aims to:
   a) Allow users full access to increase productivity
   b) Assign identical permissions to all users
   c) Disable system logging
   d) Restrict access to only what is necessary

8. SQL injection attacks target:
   a) Application logic
   b) Database queries
   c) Encryption algorithms
   d) Network routers

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9. Business continuity planning primarily focuses on:
   a) Preventing all incidents
   b) Reducing IT costs
   c) Disciplining negligent employees
   d) Ensuring essential operations continue during disruptions

10. What is network segmentation, and why is it commonly used in network security?
   a) Dividing a network into isolated segments to limit access to certain resources and improve security
   b) Combining multiple networks to allow unrestricted data sharing
   c) Creating multiple firewalls within a single network to restrict traffic
   d) Using virtual machines to simulate different network environments

11. Which of the following describes the purpose of DevSecOps?
   a) To create isolated networks for different applications
   b) To install firewalls in development environments
   c) To automate security checks, ensuring security is a shared responsibility across development, security, and operations teams
   d) To conduct regular penetration tests only after application deployment

12. In security operations, the principle of defense in depth means:
   a) Using multiple layers of security controls
   b) Increasing encryption key sizes
   c) Centralizing all access rights
   d) Eliminating redundancy

13. The primary goal of digital forensics is to:
   a) Collect, preserve, and analyze digital evidence
   b) Punish the attacker
   c) Erase infected files
   d) Enhance system performance

14. Which file system artifact helps identify when a file was last accessed?
   a) Hash value
   b) Log entries
   c) Registry key
   d) File metadata

15. Which of the following best demonstrates the concept of "users are not the enemy"?
   a) Punishing every security mistake harshly
   b) Designing systems that support secure behavior naturally
   c) Ignoring user convenience entirely
   d) Disabling user feedback options

PART 2: Short Questions [15 Marks]

Question 2: Describe two common cryptographic attacks and the defenses against them. [2 Marks]

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Question 3: How can auditing and logging contribute to maintaining OS security? [2 Marks]

Question 4: Outline key steps in hardening an operating system before deployment. [2 Marks]

Question 5: Web applications are a common attack target. Discuss three common web vulnerabilities and suggest corresponding mitigation techniques. [3 Marks]

Question 6: Explain the role of Security Information and Event Management (SIEM) in detecting and managing security incidents. [3 Marks]

Question 7: Describe the key stages of a digital forensic investigation and explain the importance of each. [3 Marks]

PART 3: Problem questions [15 Marks]

Question 8: Alice wants to use the RSA algorithm to encrypt a message. She chooses two prime numbers, p = 11 and q = 13. By using these prime numbers, compute the following components to generate the public and private keys. **[5 Marks]**

1. The modulus n
2. Euler's totient $\varphi(n)$.
3. Choose a public key exponent e
4. Compute the private key exponent d
5. Write the public and the private keys.

Question 9: Web Application Breach.

A university's online admission portal was taken offline after an attacker extracted thousands of student records using SQL injection through a poorly validated search form. The system logs revealed that the development team reused old code and skipped security testing due to tight deadlines. [7 Marks]

- Identify the main **vulnerabilities** that led to this attack.
- Suggest **technical** and **procedural measures** the university should implement to prevent similar incidents in the future.

Question 10: Insider Threat and Operational Controls.

An employee in the finance department copied sensitive payroll data onto a USB drive and leaked it to competitors. The investigation revealed that USB ports were not restricted and that data access logs were rarely reviewed. The company relied solely on trust and confidentiality agreements. **[8 Marks]**

Discuss how a combination of technical, administrative, and behavioral controls could have prevented or minimized this incident. Your answer should include:

- Identification of weaknesses
- Proposed technical, administrative, and behavioral controls that will help in preventing such attacks.

End of questions

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